Concrete Curing and Strength Estimation Guide

1 Purpose

This guide sets the rules used on our sites to decide when formwork can be removed. It relies on the maturity method, which estimates the strength of young concrete from its temperature history.

2 Strength estimation

2.1 Maturity index

The maturity index M is the sum, over the curing period, of the difference between the concrete temperature and a datum temperature, multiplied by the duration of each interval:

$$M = \sum_{i} (T_i - T_0)\,\Delta t_i$$

The datum temperature T0 is minus ten degrees Celsius for ordinary Portland cement. The index is expressed in degree-hours, and a strength of 70 percent of the design strength is reached at about 1,900 degree-hours for our standard mix.

2.2 Monitoring

Temperature records. Temperatures must be logged every thirty minutes by sensors embedded in the concrete, from the end of the pour until the specified strength is reached. Any gap longer than two hours in the record invalidates the maturity estimate of the affected pour, which must then be checked with cores tested by an accredited laboratory before the formwork is removed.

The sensor batteries last for 28 days, which covers the curing period of every mix used on our sites.

3 Minimum curing durations

When no sensor is installed, the formwork stays in place for at least the durations below.

Cement type	At 5 °C (days)	At 20 °C (days)
CEM I	6	3
CEM II	8	4
CEM III	12	6

4 Acceptance

A pour is accepted when all of the following conditions are met:

- the maturity index has reached the target of the mix
- no crack wider than 0.3 millimetres is visible
- the site engineer has signed the pour record